CIT 140 · Week 7 · Data-Quality Report

Meridian Forge · fictional inspection export v1

Learner name / date	Curtis Loop - 27 October 2026
Import status	20 rows loaded

Amber cells: your responses. Other results calculate from Prepared and Audit.

Rule	Check	Rows flagged	Share of all rows	**Affected Row_IDs**
Q01	Machine ID is required	2	10%	R03, R14
Q02	Work order is unique	2	10%	R05, R06
Q03	Status uses the agreed list	2	10%	R08, R16
Q04	Duration is in range	3	15%	R10, R18, R20
Q05	READY identifies an inspector	3	15%	R03, R07, R12
Distinct rows with any issue		11	55%	Count each row once, even when several Q-rules flag it. Do not add the five counts.

Scope: all 20 supplied records. Q02 flags every row in a repeated Work_Order_ID group; extra duplicate copies are a different measure. Unresolved fields remain flagged, not guessed.

Source: authored CIT140 case W07-S01 (Raw) and W07-S02 (rules). Report a data-quality finding, not a machine safety clearance. Explain your evidence and next action on page 2.

Week 7 · Explanation and refresh evidence

Finding: Row_ID, Q-number, actual value, source and effect

Row R03 fails two rules, Q01 and Q05 (W07-S02). Its Machine_ID cell is blank and its Inspector_ID cell is blank while the normalized Status reads READY. The record therefore states that an inspection is ready without identifying which machine was inspected or who inspected it, so nothing in the row can be traced back to a physical asset or to a person. R03 is also the reason the five rule counts total 12 while only 11 distinct rows are affected: it is counted by two rules but it is one record.

Next action: what to verify, who owns it, and how to retest

Ask the Maintenance Records Owner to verify the Machine_ID for R03 against the original inspection record for Work_Order_ID WO-1003, and ask the Shift Supervisor to verify the READY status and the inspector assignment for the same work order. Neither value is entered until the owning role confirms it. Once a confirmed correction is entered on Raw, select Data > Refresh All and retest Q01 and Q05 for R03. Both should return OK, and the distinct-row count should fall from 11 to 10.

Limit: why you cannot guess a correction or prove accuracy

I cannot supply the missing Machine_ID or Inspector_ID because the spreadsheet does not contain them and I have not seen the original inspection record. Any value I typed would be invented data that looks identical to verified data once it is saved. W07-S03 also limits what passing these rules can mean: the five checks test the record, not the machine. A row can satisfy Q01 to Q05 and still describe an inspection that was recorded incorrectly. A FLAG means a record needs checking, and an OK means the record is internally consistent, not that a machine is safe to operate.

Professional connection: one framework idea and how you used it

The Government Data Quality Framework describes data quality as a set of named dimensions rather than one overall score, separating completeness, uniqueness, validity and consistency. I used that separation to report the five checks individually instead of a single count of bad rows: Q01 and Q05 test completeness, Q02 tests uniqueness, Q03 and Q04 test validity, and Q05 also tests consistency across two fields. Keeping them apart is what makes the next action specific, because a blank Machine_ID and an out-of-range duration go to different owners and need different checks. The framework also treats accuracy as something

Refresh test: before, changed value, after, and restored result

Before the test, R01 Inspection_Minutes was 20 and Q04 flagged 3 rows: R10 (zero), R18 (75) and R20 (blank). I changed R01 Inspection_Minutes to 0 on Raw and selected Data > Refresh All. Q04 then flagged 4 rows, adding R01, because 0 falls outside the range of 1 to 60 required by W07-S02. I restored R01 to 20 and refreshed again, and Q04 returned to 3 flagged rows. The query repeated its Trim and uppercase steps automatically on each refresh, which is the point of saving the preparation in Power Query rather than editing values by hand.

Export only Report: two landscape pages, page break before row 26. Review both pages.